

Buckling $\rightarrow$ Max^m limiting load at which
or $\downarrow$ the column either buckles
or tends to buckle.
or
Critical $\rightarrow$ Buckling means the lateral
displacement during max^m loading
condition and it
depends on — length and
cross-section Area.

Euler's theory

This theory refers to the columns that are very long as compared to lateral dimensions.

- Effect of direct compression is ignored.
- buckling depends on elastic modulus (E) not on yield strength.

Assumption

- (1) Column is initially perfect straight with uniform cross-section area.
- (2) Material is homogenous and isotropic.
- (3) Fails due to buckling not direct compression.
- (4) Length is very large compared to lateral dimensions.
- (5) Loading under elastic limit (Hooke's law obeyed).
- (6) Loading is perfectly axial.
- (7) Weight of column is not considered.

Effective length

Distance b/w adjacent points of inflexion on the elastic curve.
point of inflexion is positioned near the end of the column.

End conditions

Both ends hinged
Both ends fixed
1-fixed & 1-hinged
1-fixed & 1-free

$l \leftarrow$ Actual length

Effective length

$l_e = l$
$l_e = l/2$
$l_e = l/\sqrt{2}$
$l_e = 2l$

1 Both ends Hinged

Moment at x .

$$M = -Py$$

$$\frac{d^2y}{dx^2} = \frac{M}{EI}$$

put M .

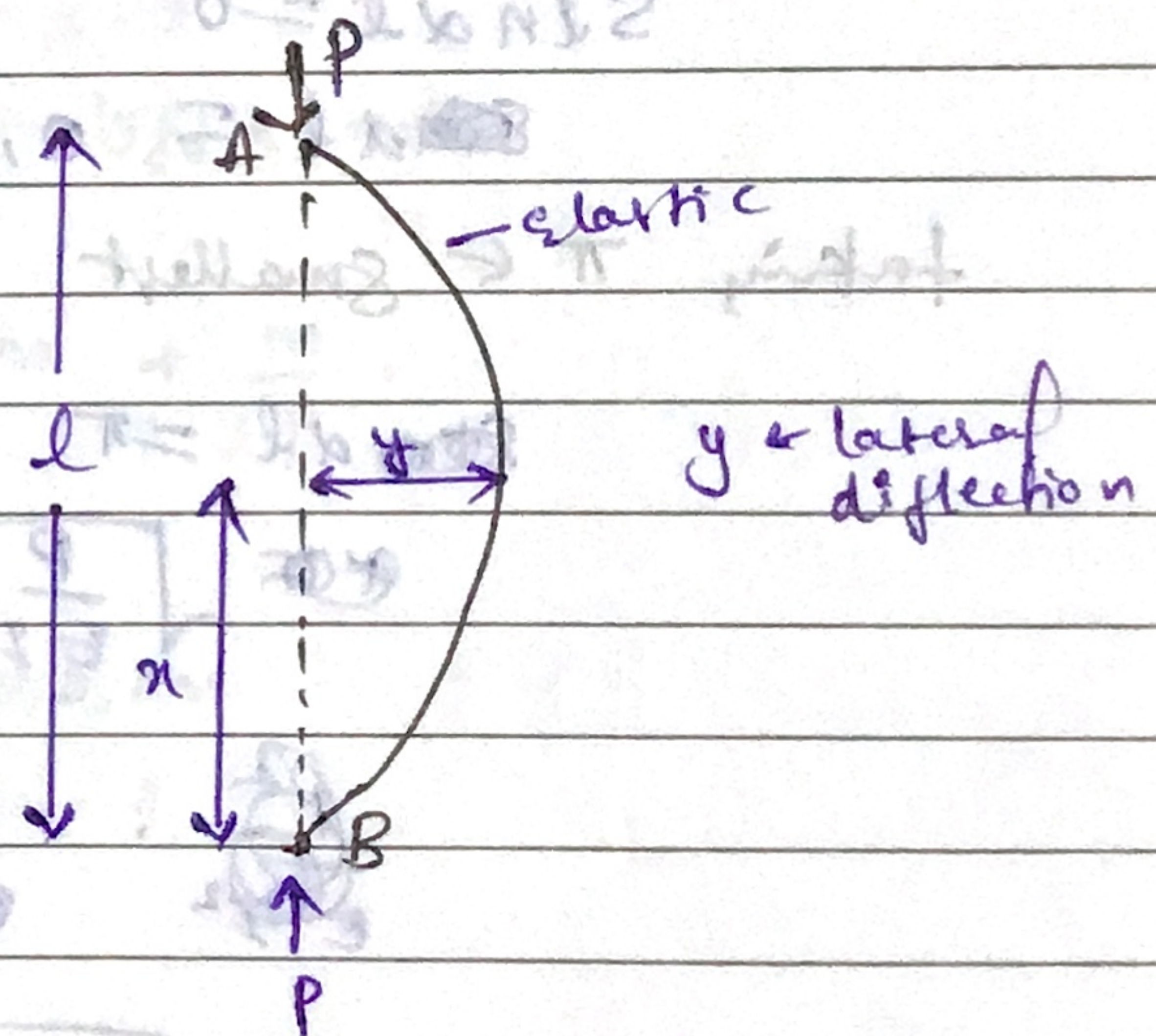
$$\frac{d^2y}{dx^2} + \frac{Py}{EI} = 0$$

let

$$\alpha = \sqrt{\frac{P}{EI}}$$

$$\frac{d^2y}{dx^2} + \alpha^2 y = 0$$

2nd order diff. eqn.



Let $y = C_1 \cos \alpha x + C_2 \sin \alpha x$.

Apply boundary conditions to find C_1 and C_2 and thus P .

(i) $x=0, y=0$

$$0 = C_1 + 0$$

$$C_1 = 0$$

(ii) $x=l, y=0$

$$0 = 0(\cos \alpha l) + C_2 \sin \alpha l$$

$$0 = C_2 \sin \alpha l$$

If C_2 is zero then there will be no deflection but there is so $C_2 \neq 0$.

So,

$$\sin \alpha l = 0$$

$$\alpha l = 0, \pi, 2\pi, \dots$$

taking π ← smallest admissible value.

$$\alpha l = \pi$$

$$\sqrt{\frac{P}{EI}} \cdot l = \pi$$

$$\frac{P}{EI} l^2 = \pi^2$$

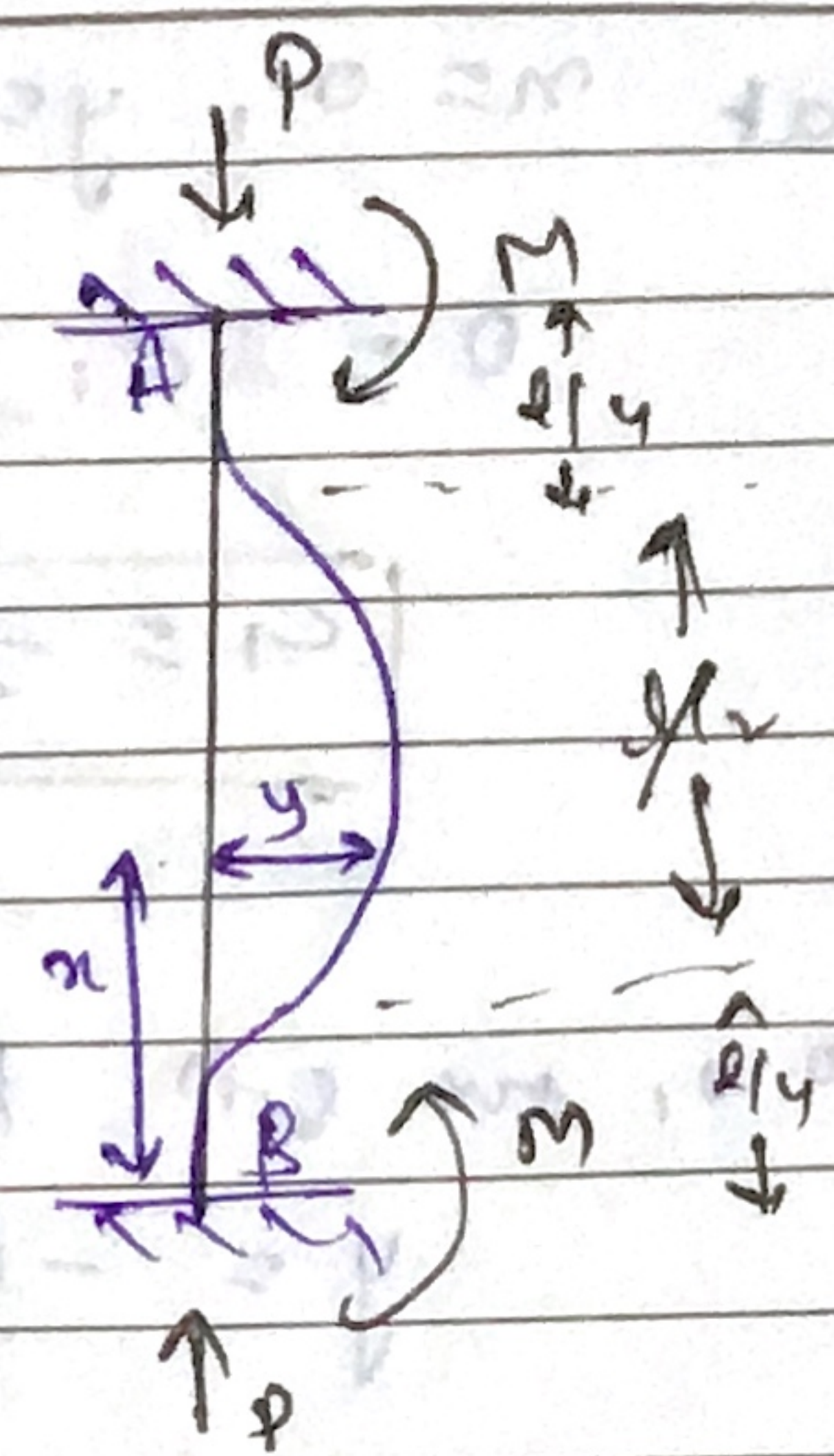
$$P = \frac{\pi^2 EI}{l^2}$$

$P_{critical}$
or
Buckling

[2] Both Ends fixed

There is restrained moment at each end say M .

moment at x
 $\rightarrow M - Py$



So

$$\frac{d^2y}{dx^2} = \frac{M - Py}{EI}$$

$$\frac{d^2y}{dx^2} + \frac{P}{EI} y = \frac{P \cdot M}{EI \cdot P}$$

$$\frac{d^2y}{dx^2} + dy = d \cdot \frac{M}{P}$$

2nd order diff. eqn.

Soln $\downarrow$

$$y = C_1 \cos dx + C_2 \sin dx + \frac{M}{P}$$

$\downarrow$ diff

$$\frac{dy}{dx} = -C_1 \sin dx + C_2 \cos dx$$

Boundary conditions

(1) at $x=0$, $\frac{dy}{dx} = 0$

$$0 = 0 + C_2$$

$$\boxed{C_2 = 0}$$

(ii) at $x=0$, $y=0$

$$0 = C_1 + 0 + \frac{M}{P}$$

$$C_1 = -\frac{M}{P}$$

Now, the eqn becomes.

$$y = -\frac{M}{P} \cos \alpha x + \frac{M}{P}$$

(iii) At $x=l$, $y=0$

$$0 = -\frac{M}{P} \cos \alpha l + \frac{M}{P}$$

$$\cos \alpha l = 1$$

$$\alpha l = 0, 2\pi, 4\pi, \dots$$

$$\sqrt{\frac{P}{EI}} \cdot l = 2\pi$$

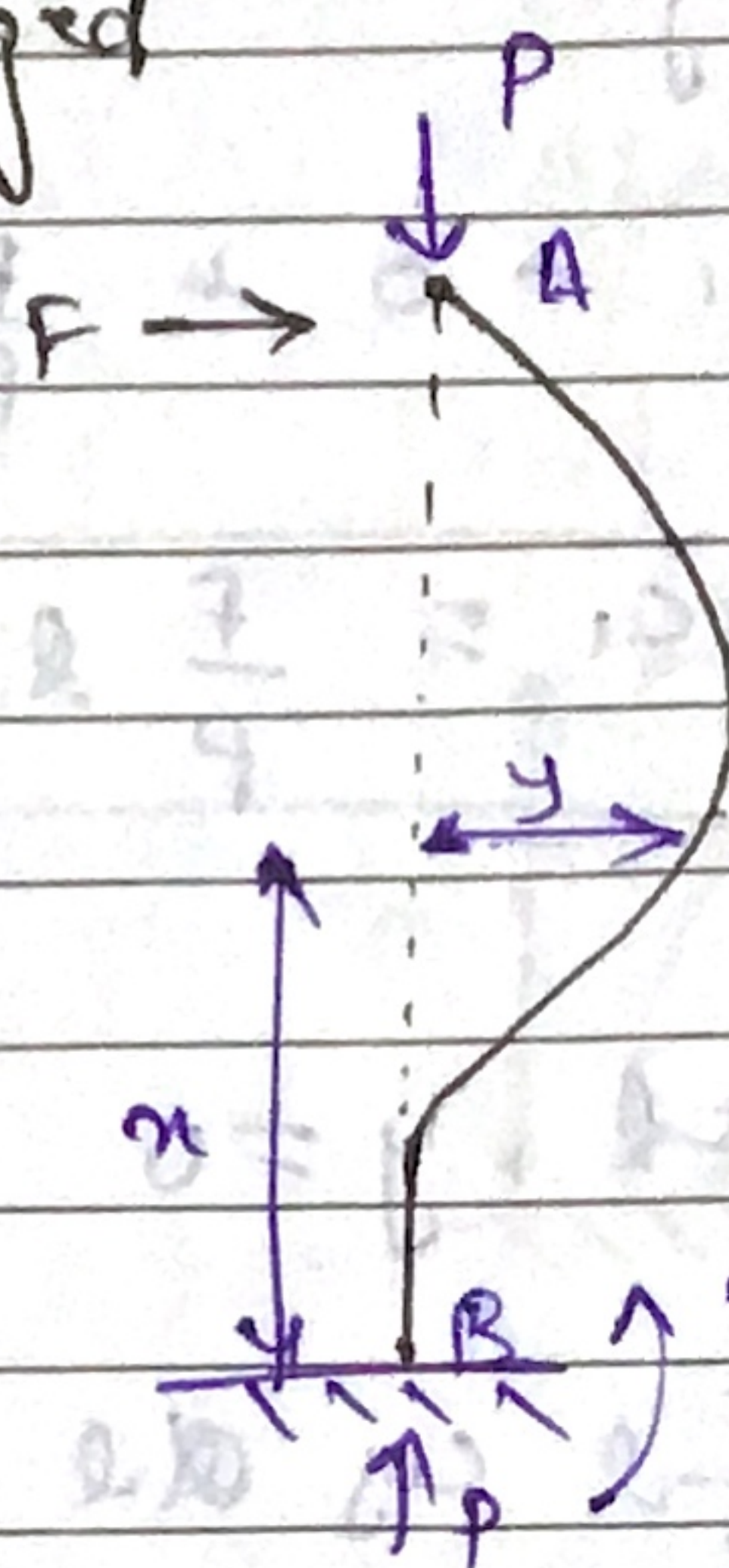
$$\frac{P}{EI} l^2 = 4\pi^2$$

$$P = \frac{4\pi^2 EI}{l^2}$$

$P_{critical}$ or
Buckling

3 One end fixed, other is hinged

M is restrained moment
to counter this another
moment is considered by
Applying F at A



moment at n is

$$= F(l-n) - Py$$

Now,

$$\frac{d^2y}{dn^2} = \frac{F(l-n) - Py}{EI}$$

$$\frac{d^2y}{dn^2} + dy = \frac{F(l-n)}{P}$$

2nd order diff eqn

Soln

$$y = C_1 \cos dn + C_2 \sin dn + \frac{F(l-n)}{P}$$

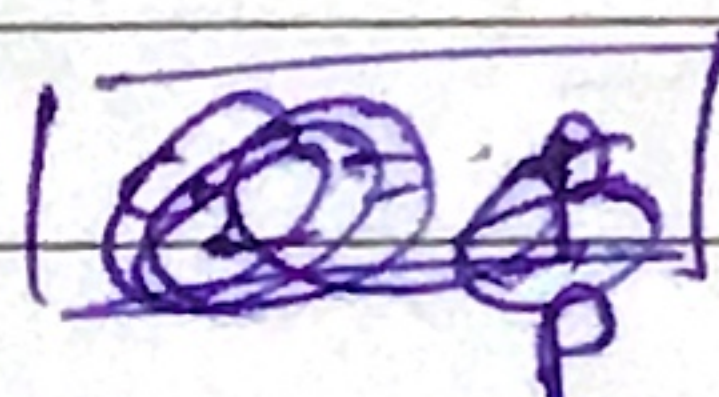
diff

$$\frac{dy}{dn} = -C_1 \sin dn + C_2 \cos dn - \frac{F}{P}$$

Boundary conditions

(i) at $n=0$, $\frac{dy}{dn} = 0$

$$0 = 0 + C_2 - \frac{F}{P}$$



$$C_2 = \frac{F}{P}$$

(ii) $x=0, y=20$

$$0 = C_1 + 0 + \frac{F}{P}(1-0)$$

$$C_1 = \frac{F}{P}$$

(iii) at $x=l, y=0$

$$0 = \frac{F}{P}l \cos \alpha l + \frac{F}{P}d \sin \alpha l - \frac{F}{P}l$$

$$l \cos \alpha l = d \sin \alpha l$$

$$\frac{d}{l} = \tan \alpha l$$

$$\tan \alpha l = \frac{d}{l}$$

$\alpha l = 0$ or 4.493 radians
 (Roman Radian)

0 is not possible.

$$\alpha l = 4.493$$

$$\sqrt{\frac{P}{EI}} \cdot l = 4.493$$

$$\frac{P}{EI} l^2 = 20.187$$

$$\frac{P}{EI} l^2 = 2\pi^2 \text{ (approx)}$$

$$P = \frac{2\pi^2 EI}{l^2}$$

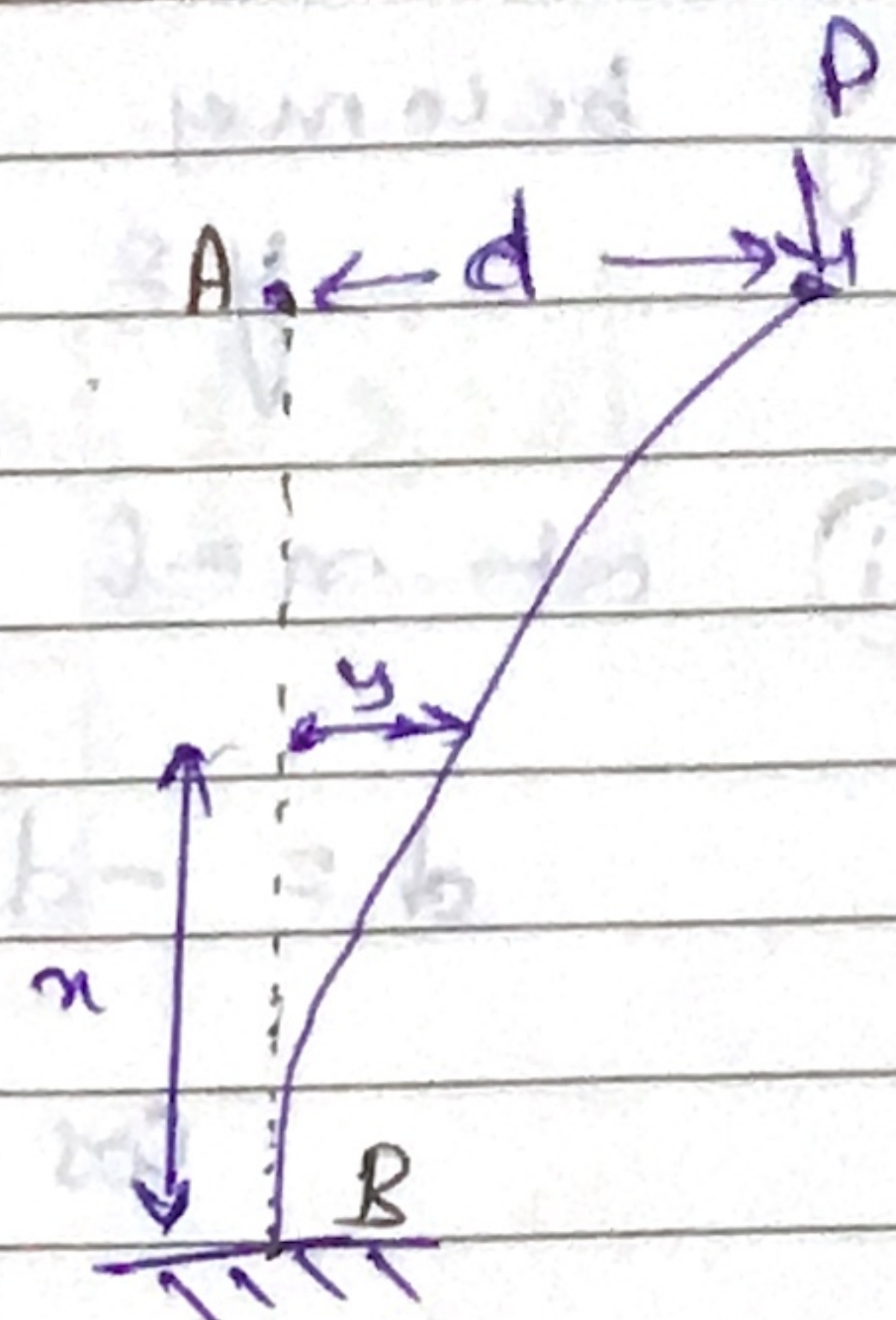
[4] One end is fixed & other is free

Moment at x is
 $P(d-y)$

So, $\frac{d^2y}{dx^2} = \frac{P(d-y)}{EI}$

$\frac{d^2y}{dx^2} + \alpha^2 y = \alpha d$

2nd diff eqⁿ



Solⁿ

$y = C_1 \cos \alpha x + C_2 \sin \alpha x + d$

↓ diff.

$\frac{dy}{dx} = -C_1 \alpha \sin \alpha x + C_2 \alpha \cos \alpha x$

Boundary condition

(i) At $x=0$, $\frac{dy}{dx} = 0$

$0 = 0 + C_2 \alpha$

$C_2 = 0$

(ii) At $x=d$, $y=0$

$0 = C_1 + 0 + d$

$C_1 = -d$

Sol^y y becomes
 $y = -d \cos \alpha x + d$

(iii) at $x=l$ $y=d$

$$d = -d \cos \alpha l + d$$

$$\cos \alpha l = 0$$

$$\alpha l = \pi/2, 3\pi/2, \dots$$

$$\alpha l = \pi/2$$

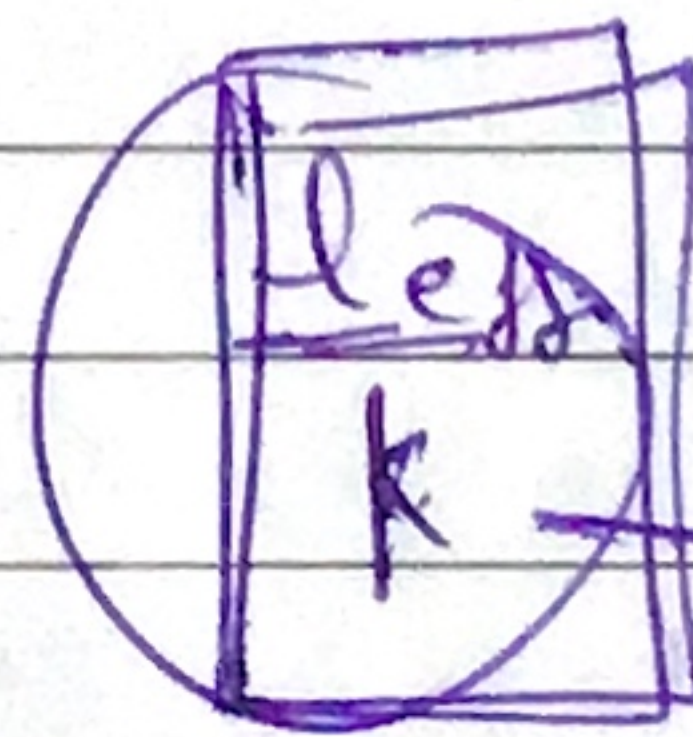
$$\sqrt{\frac{P}{EI}} \cdot l = \pi/2$$

$$\frac{P}{EI} l^2 = \frac{\pi^2}{4}$$

$$P = \frac{\pi^2 EI}{4 l^2}$$

Critical or
Buckling

Slenderness ratio.



radius of gyration

Short

Med

Long

< 30

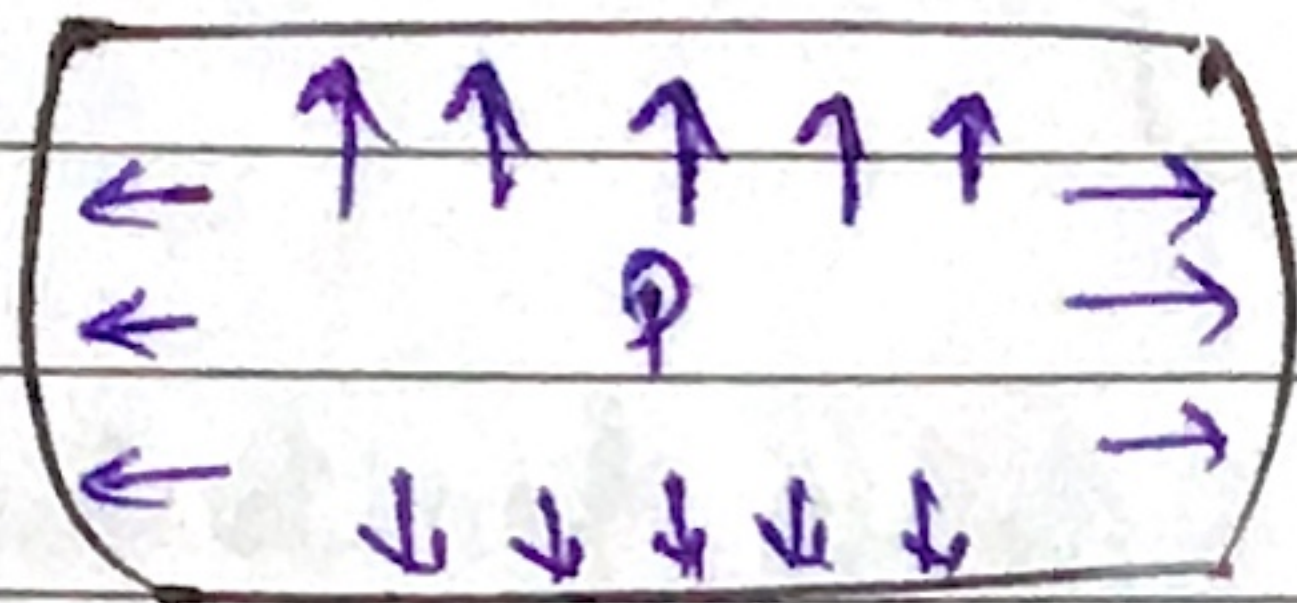
$30 - 120$

> 120

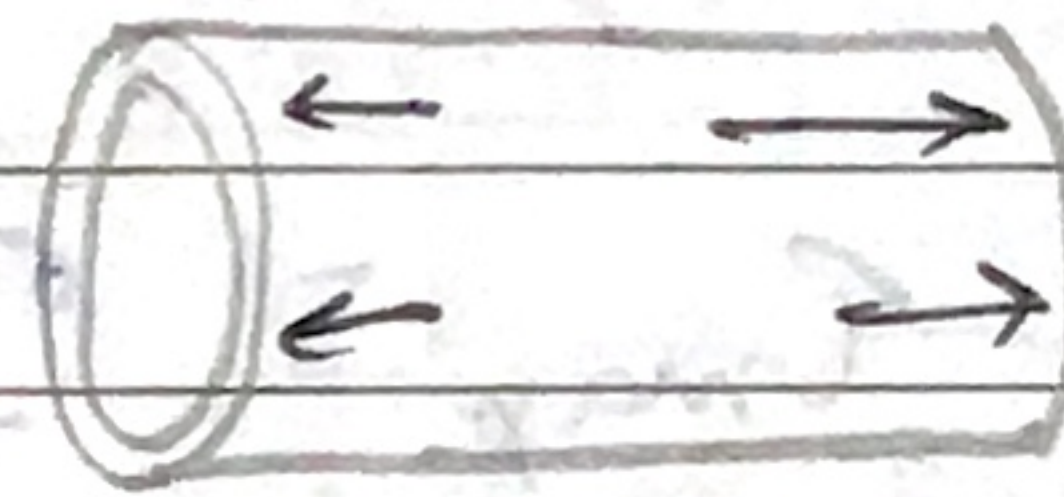
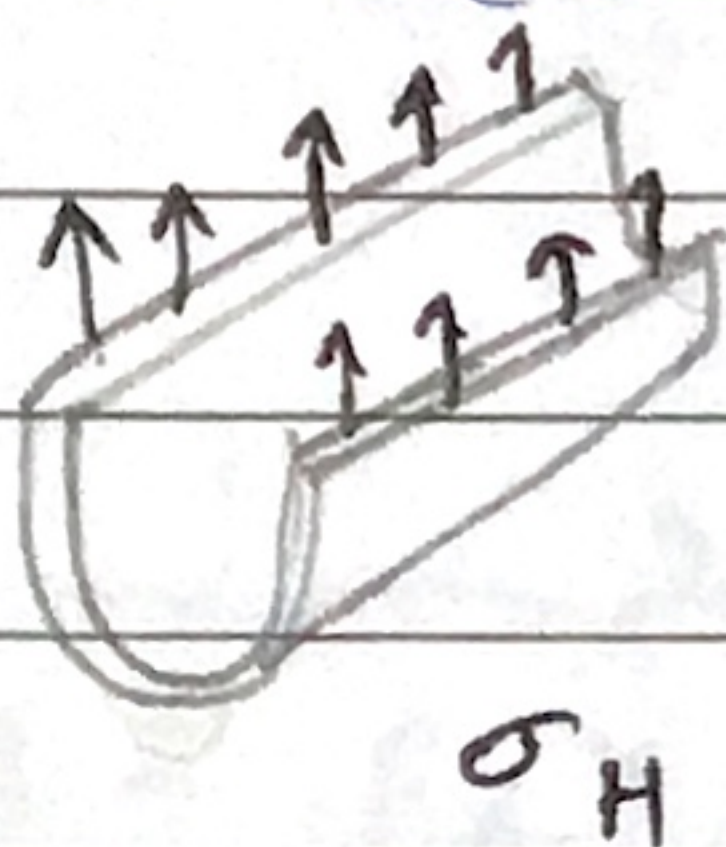
Pressure Vessels

Thin vessels $\rightarrow$ diameter or thickness $\boxed{\frac{d}{t} \geq 20}$

Thick vessels $\leftrightarrow$ $\boxed{\frac{d}{t} < 20}$



Thin cylindrical shell



Hoop or circumferential stress

Longitudinal stress.

Hoop stress.

$l \leftarrow$ length of cylinder

$t \leftarrow$ thickness

$d \leftarrow$ diameter (inside)

$p \leftarrow$ pressure due to fluid.

$$\text{Total Hoop force} = 2(\sigma_H \times t \times l)$$

$\uparrow$
Area

$$p \times \text{projected Area} = 2(\sigma_H \times t \times l)$$

$$p \times (d \times l) = 2\sigma_H t l$$

$$pd = 2\sigma_H t$$

$$\boxed{\sigma_H = \frac{pd}{2t}}$$

longitudinal stress

total longitudinal force = $\sigma_l \times \pi d \times l$

P x projection Area = $\sigma_l \times \pi d \times l$

$$P \left(\frac{\pi d^2}{4} \right) = \sigma_l \pi d l$$

$$\sigma_l = \frac{P d}{4 l}$$

Max^m shear stress

$$\tau_{max} = \frac{\sigma_H - \sigma_l}{2}$$

$$\tau_{max} = \frac{P d}{8 l}$$

$$\sigma_H = 2 \times \sigma_l$$

$$\boxed{\epsilon_{\text{strain}}}$$

Hoop strain, $\epsilon_H = \sigma_H - \mu \sigma_L$

$$= \frac{Pd}{2t} - \mu \frac{Pd}{4t}$$

$$\boxed{\epsilon_H = \frac{Pd}{4t} (2 - \mu)}$$

Longitudinal strain, $\epsilon_L = \sigma_L - \mu \sigma_H$

$$= \frac{Pd}{4t} - \mu \frac{Pd}{2t}$$

$$\boxed{\epsilon_L = \frac{Pd}{4t} (1 - 2\mu)}$$

$$\Delta L = \epsilon_L \times L$$

$$\Delta d = \epsilon_H \times d$$

Volumetric strain,

Initially $V = \frac{\pi}{4} d^2 \cdot L$

Final vol^m, $V + dV = \frac{\pi}{4} (d + \delta d)^2 \cdot (L + \delta L)$

$$dV = (V + dV) - V$$

$$dV = \frac{\pi}{4} (d + \delta d)^2 \cdot (L + \delta L) - \frac{\pi}{4} d^2 L$$

$$\frac{dV}{V} = \frac{\pi/4 \left((d + \delta d)^2 (L + \delta L) - d^2 L \right)}{\pi/4 d^2 L}$$

$$\frac{dv}{v} = \frac{(d^2 + \delta d^2 + 2d\delta d)(l + \delta l) - d^2 l}{d^2 l}$$

d

$\delta d^2 \ll$ very small,
negligible.

$$\frac{dv}{v} = \frac{(d^2 + 2d\delta d)(l + \delta l) - d^2 l}{d^2 l}$$

$$= \frac{\cancel{d^2 l} + d^2 \delta l + 2d\delta d \cdot l + \cancel{2d\delta d \delta l} - \cancel{d^2 l}}{d^2 l^2}$$

Negligible

$$\frac{dv}{v} = \frac{d^2 \delta l + 2d\delta d \cdot l}{d^2 l}$$

$$\boxed{\frac{dv}{v} = \frac{\delta l}{l} + 2 \cdot \frac{\delta d}{d}}$$

$$\boxed{\epsilon_v = \epsilon_l + 2\epsilon_d} \quad \#$$

Calculating ϵ_v

$$\epsilon_v = \frac{Pd}{4t} (1 - 2\mu) + 2 \frac{Pd}{4t} (1 - 2\mu)$$

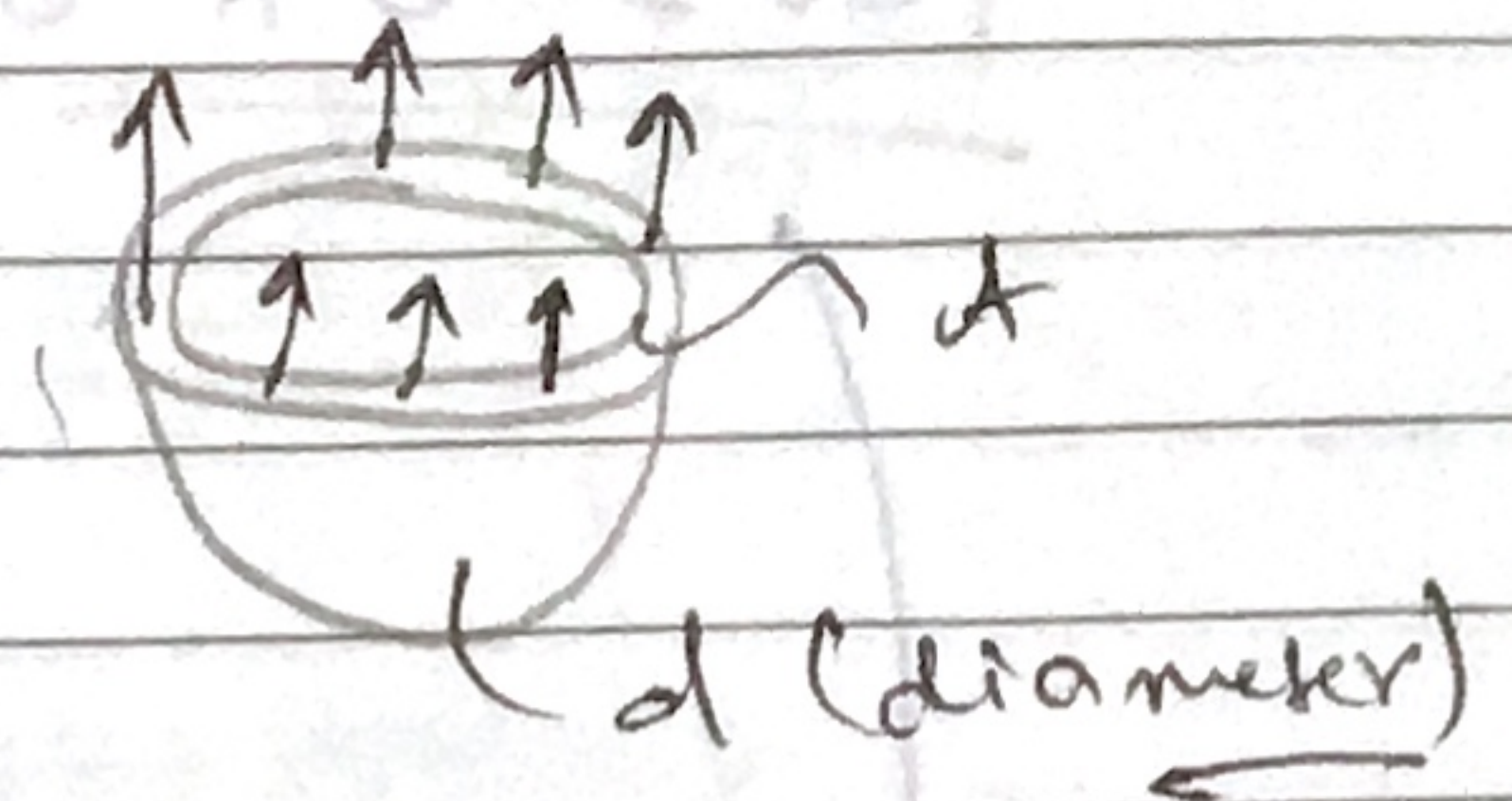
$$\epsilon_v = \frac{Pd}{4t} (1 - 2\mu + 2 - 4\mu)$$

$$\boxed{\epsilon_v = \frac{Pd}{4t} (3 - 4\mu)}$$

Ver. stress:
(cylinder)

Thin spherical shell

$$\sigma_H \times (\pi d \times t) = p \times \pi d^2$$



$$\sigma_H = \frac{p d}{4 t} \quad \text{Hoop stress}$$

By of symmetry in all directions

$$\sigma_L = \frac{p d}{4 t}$$

Volumetric Strain

Initial vol^m, $V = \frac{\pi}{6} d^3$

final vol^m, $V = \frac{\pi}{6} (d + \delta d)^3$

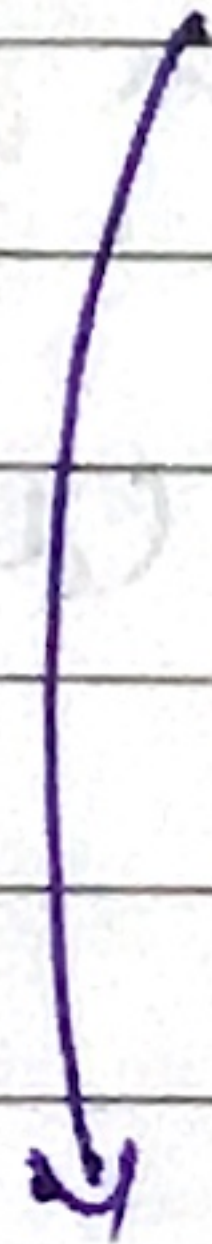
$$\Sigma_v = \frac{\text{change in vol}}{\text{initial vol}} = \frac{\frac{\pi}{6} (d + \delta d)^3 - \frac{\pi}{6} d^3}{\frac{\pi}{6} d^3}$$

$$\Sigma_v = \frac{(d^3 + \delta d^3 + 3d^2 \delta d + 3\delta d^2 d) - d^3}{d^3}$$

$$\Sigma_v = \frac{d^3 + 3d^2 \delta d - d^3}{d^3}$$

$$\Sigma_v = 3 \frac{\delta d}{d}$$

$$\boxed{\epsilon_v = 3 \times \epsilon_H} \#$$



$$\epsilon_H = \frac{\sigma_H - \mu \sigma_d}{E}$$

$$\epsilon_H = \frac{Pd}{4tE} - \mu \frac{Pd}{4tE}$$

$$\epsilon_H = \frac{Pd}{4tE} (1 - \mu)$$

$$\boxed{\epsilon_v = 3 \cdot \frac{Pd}{4tE} (1 - \mu)} \#$$

Vol. strain (sphere)

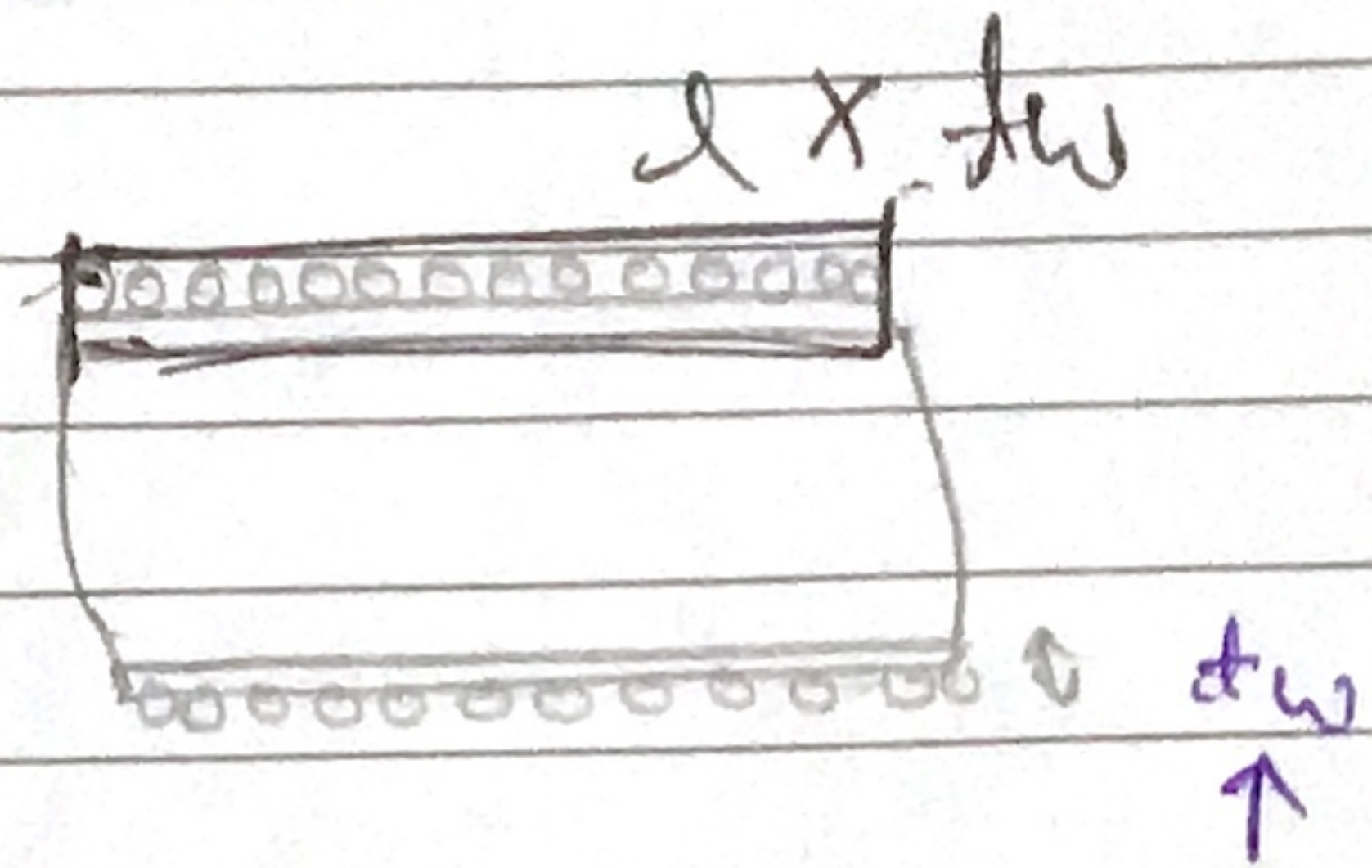
Wire wound thin cylindrical vessel

$d \leftarrow$ Dia of wire

for unit length

$$t_w \times d = \frac{\pi}{4} \cdot d^2$$

$$t_w = \frac{\pi d}{4}$$



thickness
of eq. solid shell.

Initial tensile stress of wire.

$$\sigma_w = \frac{4T}{\pi d^2}$$

$T \leftarrow$ initial force tensile

Initial hoop stress.

$$\sigma_H \times t = \sigma_w \times t_w \quad \left(\text{balance of force} \right)$$

$$\sigma_H \cdot t = \sigma_w \cdot t_w$$

$$\sigma_{wire} = \frac{F}{A}$$

$$\sigma_w = \cdot$$

$$\sigma \geq \sigma$$

$$\sigma_H = \frac{PD}{2t} = \frac{\sigma_w A_w}{A_s + A_w}$$